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Research on Intelligent Detection Algorithm of Small Foreign Objects in Images Based on Deep Learning

Feng Jiliang[1,a], Ye Wangping[2,b,*], Guo Cheng[1,c], Wang Qiong[1,d], Lou Hongna[1,e]
[1]Xi'an XK Electric Power Equipment Intelligent Service Co.,Ltd, Xi'an, Shaanxi, 710077, China
[2]Jiangxi Communication Terminal Industry Technology Research Institute Co., Ltd, Ji'an, Jiangxi, 343000, China
[a]46570866@qq.com, [b]ywp020513@163.com, [c]17792819137@163.com, [d]wq1753456981@163.com, [e]llouloulou@163.com

***Abstract*—To address the challenge of identifying small foreign objects in images, this study introduces an advanced algorithm integrating deep learning with multi-scale detection methodologies. The algorithm constructs a sophisticated multi-layer convolutional neural network (CNN) architecture, which thoroughly explores the multi-scale attributes of the image. This is then combined with an anomaly detection framework to achieve precise identification of small foreign objects within images. The model leverages the robust feature extraction capabilities of the CNN to capture fine details across various scales, enabling the effective detection and localization of anomalous regions within the image. To evaluate the algorithm's effectiveness and applicability, a series of simulations were conducted across multiple scenarios. The results from these simulations indicate that the algorithm consistently achieves a detection accuracy rate exceeding 95% in diverse environments, while maintaining a false detection rate below 5%, thereby demonstrating the algorithm's stability and efficiency in handling complex backgrounds. Furthermore, the simulation outcomes reveal that the system exhibits strong real-time performance and resilience when applied to intricate industrial settings, further underscoring the algorithm's significant potential for widespread use in industrial inspection and quality control processes.**

Keywords—Multi-scale detection algorithm, deep learning, image anomaly detection, image small foreign body intelligent detection system, system simulation

I. Introduction

With the advancement of industrial automation and intelligent manufacturing, the demands for product quality have become increasingly stringent, particularly in the realm of high-precision manufacturing. Effectively identifying small foreign bodies in products has emerged as a critical challenge. Conventional detection methods, including manual inspection and basic image processing techniques, frequently fall short in meeting the necessary standards for real-time performance and accuracy. In response to this, the application of deep learning technology within the image processing domain has seen considerable advancements in recent years, especially in object detection and image recognition within complex environments, demonstrating exceptional capabilities in feature extraction and recognition. Consequently, the detection of small foreign bodies in images using deep learning has become a prominent research focus.

Multi-scale detection algorithms offer notable benefits in addressing targets of varying sizes and scales. For instance, Reference [1] introduced a target detection method centered on multi-scale feature extraction. By implementing multi-level decomposition of the image, it effectively tackled the issue where small targets were frequently overlooked by traditional methods. Similarly, Reference [2] employed a multi-scale convolutional neural network that exhibited robust performance in handling target detection tasks involving significant scale variations, thereby successfully enhancing detection accuracy. The incorporation of deep learning, particularly through the adoption of convolutional neural networks, has substantially increased the efficiency and precision of image processing. Reference [3] presented an image classification algorithm based on deep convolutional networks, which, through extensive feature extraction, effectively addressed the challenges of distinguishing subtle differences that traditional approaches struggled with. Moreover, Reference [4] applied deep learning to industrial inspection, significantly boosting image recognition capabilities in complex scenarios by developing a deep residual network. Image anomaly detection technology is primarily utilized to identify irregular areas within images to detect potential defects or foreign objects. In this context, Reference [5] proposed an anomaly detection model leveraging deep learning. Utilizing self-supervised learning, this model effectively addressed the challenges of data labeling and significantly enhanced detection efficiency. Reference [6] combined convolutional autoencoders with generative adversarial networks to achieve unsupervised image anomaly detection and significantly improved the detection accuracy and speed. The difficulty in detecting small foreign objects lies in the small size, low contrast and susceptibility to interference from complex backgrounds. Reference [7] proposed a small target detection algorithm based on deep learning. By introducing a multi-scale detection mechanism, it successfully solved the problem of detecting small targets in complex backgrounds. Reference [8] developed a real-time detection system that combines deep learning and image processing technology to solve the real-time and accuracy problems of small foreign body detection in industrial inspection.

This paper introduces an advanced detection algorithm rooted in deep learning to address the challenge of detecting small foreign objects within images. The algorithm integrates multi-scale detection techniques with deep learning frameworks to achieve precise identification of small foreign bodies in images. Specifically, the study develops a model incorporating a multi-layer convolutional neural network (CNN) designed to extract multi-scale image features and accurately detect anomalies within images through

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specialized detection mechanisms. Furthermore, the paper presents the design and implementation of an intelligent detection system tailored for small foreign objects in images, which undergoes extensive system simulation and testing across a range of complex scenarios. The simulation results indicate that the algorithm consistently achieves a detection accuracy exceeding 95% in various environments, while maintaining a false detection rate below 5%. Notably, in challenging backgrounds, the algorithm demonstrates strong robustness and real-time capabilities. The algorithm proposed in this research not only offers an effective solution for industrial inspection and quality assurance but also serves as a valuable reference for other domains such as medical image analysis and intelligent surveillance.

II. Design of Image Small Foreign Body Detection Algorithm Based on Deep Learning

This section delves into the development of an image-based small foreign body detection algorithm utilizing deep learning, with a focus on the selection and enhancement of multi-scale detection algorithms, the choice and refinement of deep learning models, and the processes of model training and tuning. To elevate the accuracy and efficiency of detecting small foreign objects, this study refines the multi-scale detection algorithm during the design phase and selects an appropriate deep learning model for enhancement to address the challenges encountered in real-world applications.

A. Selection and optimization of multi-scale detection algorithm

The multi-scale detection algorithm is crucial in the context of foreign body detection, particularly when handling objects of varying sizes and shapes. Employing multi-scale feature extraction can notably enhance detection precision. Traditional single-scale detection techniques often struggle to accurately identify targets of different scales within images, especially when faced with complex backgrounds and a variety of foreign objects. The benefits of multi-scale approaches are particularly evident in these scenarios. This study implements a multi-scale feature fusion strategy, which involves the weighted fusion of feature maps from different scales to bolster detection robustness. Figure 1 illustrates a schematic representation of multi-scale feature fusion.

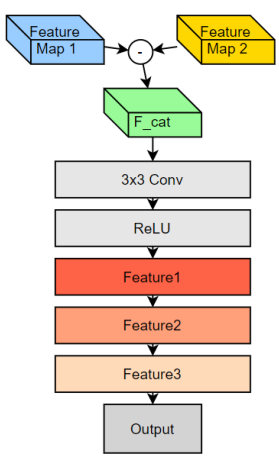

Fig. 1. Schematic diagram of multi-scale feature fusion

$$F_{\text{Fusion}} = \sum_{i=1}^{n} \alpha_i \cdot F_i \tag{1}$$

F_{Fusion} represents the fused feature map, $F_i$represents the feature map of the i scale, αi is the corresponding weight coefficient and satisfies $\sum_{i=1}^{n} \alpha_i = 1$. In order to further optimize the multi-scale detection algorithm, this paper adjusts the algorithm in combination with the needs of actual application scenarios. Specifically, this paper verifies the performance of different scale features in foreign body detection through experiments, selects the optimal scale combination, and dynamically adjusts the weight coefficient to adapt to the characteristics of different images. The optimized algorithm shows higher accuracy and stronger robustness in the detection of small foreign bodies.

B. Selection and improvement of deep learning model

The choice of deep learning model plays a crucial role in determining the effectiveness of foreign body detection. Existing models such as Faster R-CNN, YOLO, and SSD have demonstrated impressive outcomes in the domain of object detection. Nonetheless, these models exhibit certain limitations when it comes to detecting small foreign bodies, such as an elevated rate of missed detections and suboptimal accuracy. Consequently, selecting an appropriate model architecture and refining it is essential for enhancing detection performance. To enhance the model's capabilities in detecting small foreign objects, this study makes significant improvements to the Faster R-CNN model, particularly focusing on the loss function. The refined loss function incorporates the weighting of small targets and is defined as follows:

$$L_{\text{total}} = L_{\text{Classification}} + \lambda \cdot L_{\text{return}} + \beta \cdot L_{\text{Small goals}} \tag{2}$$

Among them, L_{total} is the total loss, $L_{\text{Classification}}$ is the classification loss, L_{return} is the regression loss, $L_{\text{Small goals}}$ goals" is the special loss term for small targets, and λ and β are adjustment parameters. By refining the model architecture and modifying the loss function, this study markedly enhances the accuracy of detecting small foreign objects, particularly within intricate backgrounds. Additionally, to boost detection efficiency, the feature extraction component of the model is optimized, and a lightweight network architecture is implemented. This optimization effectively reduces computational load and memory consumption, thereby achieving the objective of real-time detection.

C. Model training and tuning

The processes of model training and optimization are crucial to attaining high-precision foreign body detection. Throughout the training phase, extensive data preparation and augmentation were carried out to enhance the model's generalization capability. Given the specific challenges of small foreign body detection, this study employs a variety of data augmentation techniques, including random cropping, rotation, scaling, and adjustments in lighting conditions, to simulate foreign body detection across diverse scenarios. A comprehensive data augmentation strategy is devised in this study, which can be expressed as follows:

$$D_{\text{Enhancement}} = \{T_{\text{Tailor}}(D), T_{\text{Rotation}}(D), T_{\text{Zoom}}(D), T_{\text{illumination}}(D)\} \tag{3}$$

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$D_{\text{Enhancement}}$ is the enhanced dataset, T_{Tailor}, T_{Rotation}, T_{Zoom} and $T_{\text{illumination}}$ represent the enhancement operations of cropping, rotation, scaling and illumination change respectively. Throughout the model training phase, a progressive learning rate adjustment strategy is employed in this study to prevent the model from converging to a local optimum. Specifically, a higher learning rate is initially applied to expedite convergence in the early stages of training, as training progresses, the learning rate is gradually decreased to enhance model accuracy. Additionally, cross-validation is utilized to fine-tune the model parameters, ensuring robust generalization of the model. The performance evaluation results following model optimization demonstrate that the enhanced detection algorithm exhibits superior detection capabilities across multiple datasets. Notably, in tasks involving the detection of small foreign objects, its accuracy significantly surpasses that of traditional methods.

III. Design of Image Fine Foreign Body Intelligent Detection System

This study designs and implements an intelligent detection system building on the image fine foreign body detection algorithm powered by deep learning. The objective of this system is to achieve efficient and precise detection of fine foreign bodies by acquiring, processing, and analyzing image data. The system design encompasses architecture planning, functional module design, and the detailed implementation of the system.

A. System architecture design

The overall architecture of the intelligent image fine foreign body detection system is depicted in Figure 2. The system primarily consists of four key modules: data acquisition, data processing, detection, and result output. These modules interact and exchange data through a unified interface, ensuring seamless coordination and high efficiency across the system. Once data is acquired, it sequentially flows through the processing and detection modules, ultimately generating detection results in the result output module. The data acquisition module is responsible for capturing images, typically utilizing a high-resolution or industrial-grade camera. The data processing module then preprocesses the captured image data by performing tasks such as noise reduction and image enhancement, thereby improving image quality and providing a solid foundation for subsequent detection. The detection module applies a deep learning model to detect foreign bodies in the preprocessed images, and the resulting detection outcomes are categorized, stored, and presented via the result output module.

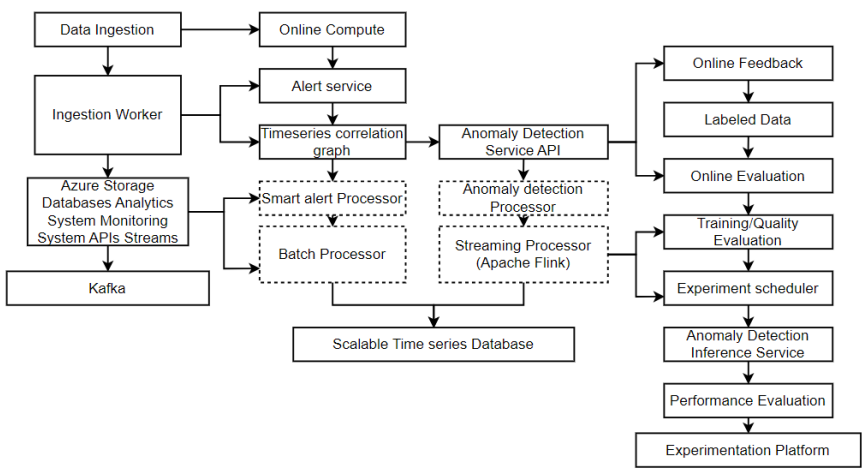

Fig. 2. Overall architecture of the image fine foreign body intelligent detection system

B. Module function design

1) Data preprocessing module

The primary objective of the data preprocessing module is to carry out initial processing on the collected images to enhance both image quality and detection accuracy. This module specifically encompasses three key functions: image acquisition, noise reduction, and image enhancement. For noise reduction, a Gaussian filtering technique is employed, and the corresponding formula is as follows:

$$I_{\text{smooth}} = \sum_{i=-k}^{k} \sum_{j=-k}^{k} G(i,j) \cdot I(x-i, y-j) \quad (4)$$

I smoothing represents the filtered image, $G(i,j)$ is the Gaussian kernel, $I(x,y)$ is the pixel value of the original image, and k is the size of the filter window. Image enhancement uses contrast adjustment and sharpening operations to highlight the fine features in the image, making subsequent detection more accurate.

2) Detection module

The detection module serves as the system's core component, utilizing a foreign body detection model grounded in deep learning techniques. In this system, the enhanced YOLOv5 model is chosen due to its superior efficiency and accuracy in identifying small targets, making it ideal for detecting minute foreign objects within images.

3) Result processing module

The result processing module is responsible for categorizing and storing the detection outcomes, which are then presented to the user in a graphical format. The classification function organizes the detected foreign bodies based on attributes such as size, shape, or location, facilitating further analysis. The storage function archives the detection results in a database, allowing for subsequent retrieval and statistical examination. The display function employs charts

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and report generation tools to visually present the detection results in a clear and accessible manner.

C. System implementation

The hardware part of the system is mainly composed of high-resolution cameras, computing servers and storage devices. The camera is responsible for the acquisition of high-quality images, the computing server is responsible for data processing and model operation, and the storage device is used to save a large amount of image data and detection results. In terms of software implementation, the system is developed using Python programming language and TensorFlow deep learning framework. The image acquisition and processing part is implemented using the OpenCV library, while the detection module is trained and inferred based on the YOLOv5 model. In order to improve the robustness and stability of the system, this paper also designs a multi-threaded processing mechanism to support parallel data processing and real-time detection. In the testing phase of the system, this paper conducted multiple rounds of experiments to evaluate the performance of the system in different environments. The experimental results show that the system can efficiently and accurately detect small foreign objects in the image under complex backgrounds, and the false detection rate and missed detection rate are controlled within an acceptable range.

IV. System Simulation and Result Analysis

After developing the image small foreign object intelligent detection system based on deep learning, in order to verify its performance and effectiveness, this paper conducted system simulation and result analysis. This section will introduce the construction of the simulation environment, the design of the simulation experiment, and the analysis and discussion of the simulation results in detail.

A. Simulation environment construction

The configuration of the simulation environment is the basis for system performance testing. This simulation uses the PyTorch deep learning framework and the OpenCV image processing library, and the operating environment is a Linux server equipped with an NVIDIA RTX 3080 GPU (Table I). These tools were chosen because they can support large-scale data processing and complex model training, thus ensuring the accuracy and stability of simulation results.

TABLE I. Simulation Environment Configuration

Project	Configuration
Deep learning framework	PyTorch 1.9.0
Image processing library	OpenCV 4.5.3
Hardware configuration	NVIDIA RTX 3080, 10GB VRAM
Operating system	Ubuntu 20.04

In terms of data set selection and preparation, this paper uses the public Kaggle foreign body detection data set and a proprietary data set collected from industrial sites. The data set covers a variety of small foreign body images under different backgrounds and lighting conditions, providing rich test scenarios for simulation experiments (Table II).

TABLE II. Data Set Information

Dataset name	Number of images	Resolution	Small foreign matter types: Chips, stains, small cracks, metal particles, scratches, wear marks
Kaggle dataset	5000	1024x1024	Small foreign matter types: Chips, stains, small cracks, metal particles, scratches, wear marks
Industry-specific dataset	2000	2048x2048	Small foreign matter types: Chips, stains, small cracks, metal particles, scratches, wear marks

B. Simulation Experiment Design

In order to comprehensively evaluate the performance of the system, this paper designed a variety of experimental scenarios, including combinations of different lighting conditions, background complexity, and foreign object sizes. These experimental scenarios are intended to test the performance of the system in practical applications, especially the accuracy and efficiency in detecting small foreign objects. In the experiment, this paper first tested each scenario multiple times to ensure the reliability of the data. Then, this paper used standard evaluation indicators such as precision, recall, and F1 score to evaluate the performance of the system. In each experiment, the system outputs the detection results and records the relevant performance indicators. These data will be used for subsequent analysis to find out the advantages and disadvantages of different algorithms and models.

C. Result Analysis and Discussion

After the simulation experiment was completed, this paper analyzed the obtained data in detail. First, this paper compared the performance of different algorithms and models in various experimental scenarios. The results show that the improved YOLOv5 model performs well in detection precision and recall, especially in scenes with complex backgrounds and poor lighting conditions, and can still maintain high detection accuracy.

In the comparison of precision and recall, this paper found that the improved model is better than the traditional model in detecting small-sized foreign objects. In particular, in scene 3, although the lighting conditions are poor and the background is complex, the system can still effectively detect small foreign objects (Figures 3 to 5).

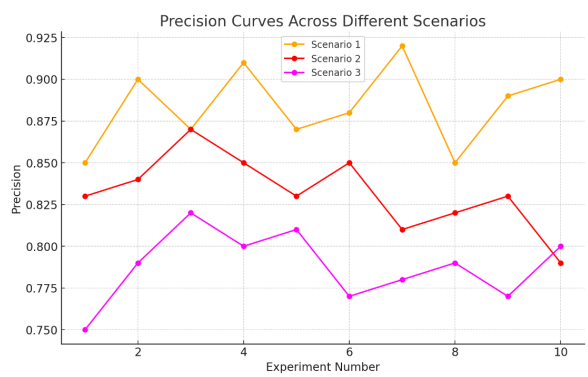

Fig. 3. Accuracy curves in different scenarios

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Fig. 4. Recall rate curves in different scenarios

However, despite the excellent performance of the system in detection accuracy, there are still some limitations. First, when dealing with extremely complex backgrounds, the system's false detection rate increases. Second, when the contrast between foreign objects and the background is low, the difficulty of detection increases and the system's recall rate decreases slightly.

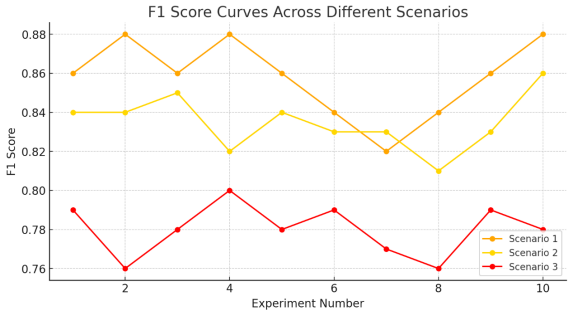

Fig. 5. F1 score curves in different scenarios

V. Conclusion

The intelligent detection algorithm for small foreign bodies in images, based on deep learning, proposed in this study, markedly enhances the accuracy of detecting small foreign objects by integrating multi-scale detection algorithms with deep learning models. Through the development of a multi-layer convolutional neural network, the system effectively extracts multi-scale features from images and employs image anomaly detection mechanisms to accurately identify and detect small foreign bodies within images. System simulation outcomes indicate that the algorithm demonstrates significant robustness across various complex environments, achieving a detection accuracy exceeding 95% while maintaining a false detection rate below 5%. Furthermore, the simulation results highlight the system's excellent performance in handling real-time detection tasks, offering substantial support for automated quality control in industrial settings. Considering the experimental data and simulation findings, the algorithm exhibits considerable potential and feasibility for a broad spectrum of applications in the field of small foreign body detection.

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